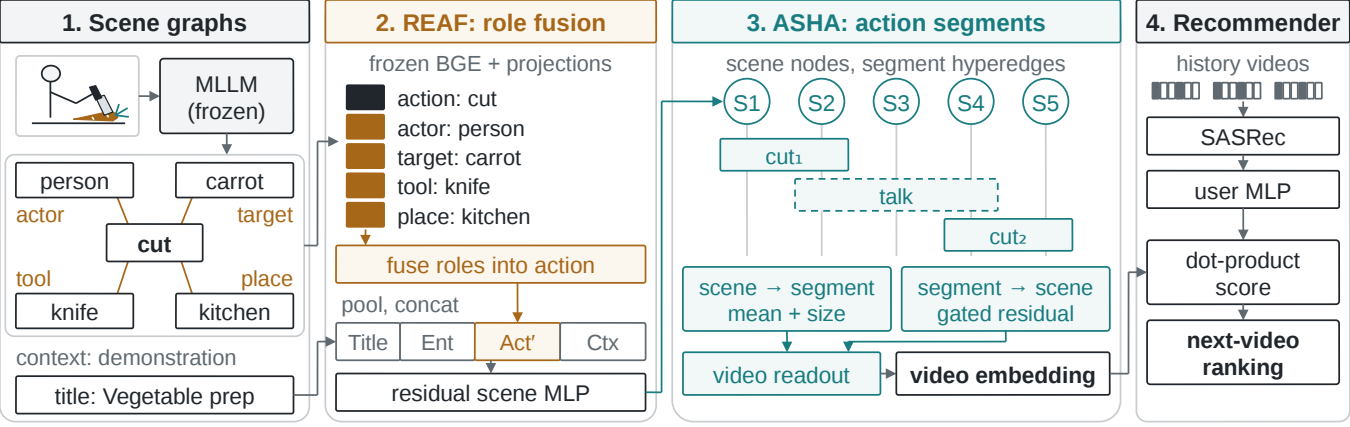
1. Scene graphs
MLLM (frozen)
person
carrot
actor
target
cut
tool
place
knife
kitchen
context: demonstration
title: Vegetable prep
2. REAF: role fusion
frozen BGE + projections
action: cut
actor: person
target: carrot
tool: knife
place: kitchen
fuse roles into action
pool, concat
Title
Ent
Act′
Ctx
residual scene MLP
3. ASHA: action segments
scene nodes, segment hyperedges
S1
S2
S3
S4
S5
cut1
talk
cut2
scene → segment mean + size
segment → scene gated residual
video readout
video embedding
4. Recommender
history videos
SASRec
user MLP
dot-product score
next-video ranking